

JEE MAIN REVISION MASTER: COMMUNICATION SYSTEMS 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 2 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Amplitude Modulation & Sideband Analysis	1 PYQs	Modulation
Standard	EM Wave Propagation & Space Communication	1 PYQs	Wave Propagation

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Amplitude Modulation & Sideband Analysis	$\mu = \frac{A_m}{A_c} = \frac{A_{\text{max}} - A_{\text{min}}}{A_{\text{max}} + A_{\text{min}}}, \quad \text{Bandwidth} = 2f_m, \quad P_t = P_c \left(1 + \frac{\mu^2}{2}\right)$
EM Wave Propagation & Space Communication	$d_{\text{max}} = \sqrt{2R_h T} + \sqrt{2R_h R}, \quad f_c = 9\sqrt{N_{\text{max}}}, \quad f_{\text{MUF}} = f_c \sec \theta_i, \quad \text{Antenna length} \geq \frac{\lambda}{4}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Amplitude Modulation & Sideband Analysis (Standard Yield — 1 PYQs)

Core Principle: Principles of amplitude modulation, carrier and sideband components ($f_c \pm f_m$), modulation index, and power relationships.

■ **Caution / JEE Pitfall:** Modulation index μ must satisfy $\mu \leq 1$ (100%) to prevent envelope distortion.

■ **Speed Trick:** Carrier power does not depend on modulation index μ , only sideband power increases as μ increases.

Representative Source PYQs:

• **[2017 Shift] Q#2 (MCQ):** "A basic communication system consists of (i) Transmitter (ii) Information sources (iii) User of information (iv) Channel (v) Receiver..." **[Ans: N/A]**

3.2 EM Wave Propagation & Space Communication (Standard Yield — 1 PYQs)

Core Principle: Propagation modes of electromagnetic waves (ground, sky, and space waves), critical frequency, MUF, and antenna LOS range calculations.

■■ **Caution / JEE Pitfall:** Ensure earth radius $R \approx 6400$ km and heights h_T, h_R are converted into the exact same units before square rooting.

■ **Speed Trick:** For an antenna of height h , maximum range is $d = \sqrt{2Rh}$, and population covered is $\pi d^2 \times \rho$.

Representative Source PYQs:

• [2018 Shift] Q#1 (MCQ): "The radius of the Earth is 6400 km . If the height of an antenna is 500 m , then its range is..."

[Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2018 (Q#1)

The radius of the Earth is 6400 km . If the height of an antenna is 500 m , then its range is

- (A) 800 km
- (B) 100 km
- (C) 80 km
- (D) 10 km

Correct Answer: None

Solution Analysis: To find the range of the antenna (i.e., the distance to the horizon), we can use the formula: $d = \sqrt{2R \bullet \blacksquare}$ where: • R is the radius of the Earth • $\blacksquare$ is the height of the antenna Since the Earth's radius is given as 6400 km and the antenna's height is 500 m, we first convert the height into kilometers: • $500\text{m} = 0.5\text{km}$ Now, plug the values into the formula: $d = \sqrt{2 \times 6400 \times 0.5} = \sqrt{6400} = 80\text{km}$ Thus, the range of the antenna is 80 km, which corresponds to Option C.